Antiallergic activities of pigmented rice bran extracts in cell assays.

Using a panel of chemical, biochemical, and cell assays, we determined inhibitory effects of extracts of the pigmented black rice brans on in vitro allergic reactions. Ethanol-water (70% v/v) extracts from 5 pigmented brans were found to be more effective than an extract from a nonpigmented rice cultivar in suppressing the release of histamine and beta-hexosaminidase from basophilic RBL-2H3 cells stimulated with both Ionophore A23187 and immunoglobulin E (IgE)-antigen complexes. Suppression was also obtained with A23187-stimulated rat peritoneal mast cells. The extent of inhibition of these 2 markers of the immune response was accompanied by an influx of calcium ions. The inhibition of the immune process by the pigmented brans was confirmed by the observed modulation of the proinflammatory cytokine gene expressions and cytokine release, as indicated by the reduction in tumor necrosis factor (TNF)-alpha, interleukin (IL)-1beta, IL-4, and IL-6 mRNA expressions determined with the reverse transcription-polymerase chain reaction (RT-PCR). Reduction of TNF-alpha, IL-1beta, and IL-6 protein release from both the cultured cell line and peritoneal cells was further confirmed by enzyme-linked immunoadsorbent assays. Rice bran from the LK1-3-6-12-1-1 cultivar was the most effective inhibitor in all assays. This particular rice variety merits further evaluation as part of a human diet to ascertain its potential to protect against allergic diseases such as hay fever and asthma.